A reflexive mere relation implying identity makes its type a set, and is equivalent to its identity type

The Univalent Foundations Program

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Editorial note. The counted statement is Theorem 7.2.2, the author's 'thm:h-set-refrel-in-paths-sets' (source lines 265-269). Its proof environment begins on the next line after the statement, so the pairing is by adjacency in the original file.

Editorial note. Both proofs written by the author are retained, since the source presents them as two proofs of the same theorem; the retained span is the whole proof environment, not a selection.

Editorial note. The retained counter values (chapter 7, section 2, theorem offset 1) restore the book's own number 7.2.2 for the counted theorem. The chapter and section headings themselves are not part of the retained context, so only the counter values, not the headings, are reproduced.

Editorial note. The context paragraph is retained because it introduces the theorem by name and points back to the earlier non-set example; no line inside the retained spans contains a citation, a figure, an included file or a non-ASCII character, so no bibliography and no asset is declared.

Editorial scope. One original result of the HoTT Book, counted once: A reflexive mere relation implying identity makes its type a set, and is equivalent to its identity type.

Allowed standard prerequisites (editorial).

- Mere propositions, sets and Axiom K. A type is a mere proposition when all its elements are equal; a type X is a set when p = q for all x,y : X and p,q : x = y; Axiom K says p = refl_x for every p : x = x, and the book proves K equivalent to being a set at Theorem 7.2.1.. The theorem is stated for a mere relation R implying identity and its conclusion is that X is a set, so both the hypothesis and the equivalence used at source lines 274-278 are level-0 material.
- Identity types, transport and dependent paths. Transport along p : x = y in a type family, ap of a dependent function, and the dependent path type; the book's Lemma 2.9.6 converts a dependent path in an arrow type into a path between transports.. The first author proof at source lines 287-296 is a computation with ap of the dependent function f(x) : R(x,x) -> (x = x) and with the arrow-type dependent path lemma.

- Total spaces, fibres and the sigma path characterization. For f : A -> B, f is an equivalence as soon as the induced map of total spaces is an equivalence (the book's Theorem 4.7.7), and paths in a sigma type are pairs of a path and a dependent path (the book's Theorem 2.7.2).. The second author proof at source lines 299-312 needs exactly these two ingredients to show each f(x,y) is an equivalence.

Source reference. Original theorem 3.1.9: sources/hott-hlevels/book-labelnumbers-first-edition.tex, source lines 1911–1911. Pointer to the earlier result quoted in the retained context paragraph.

Source reference. Original lemma 3.3.4: sources/hott-hlevels/book-labelnumbers-first-edition.tex, source lines 1863–1863. Original result cited by the retained proof but located outside the retained span. It is a navigation pointer: the counted theorem's own author proof is retained line for line, and no part of it is replaced by this reference.

Source reference. Original lemma 2.9.6: sources/hott-hlevels/book-labelnumbers-first-edition.tex, source lines 1764–1764. Original result cited by the retained proof but located outside the retained span. It is a navigation pointer: the counted theorem's own author proof is retained line for line, and no part of it is replaced by this reference.

Source reference. Original theorem 4.7.7: sources/hott-hlevels/book-labelnumbers-first-edition.tex, source lines 1891–1891. Original result cited by the retained proof but located outside the retained span. It is a navigation pointer: the counted theorem's own author proof is retained line for line, and no part of it is replaced by this reference.

Source reference. Original lemma 3.11.8: sources/hott-hlevels/book-labelnumbers-first-edition.tex, source lines 1760–1760. Original result cited by the retained proof but located outside the retained span. It is a navigation pointer: the counted theorem's own author proof is retained line for line, and no part of it is replaced by this reference.

Source reference. Original theorem 2.7.2: sources/hott-hlevels/book-labelnumbers-first-edition.tex, source lines 1853–1853. Original result cited by the retained proof but located outside the retained span. It is a navigation pointer: the counted theorem's own author proof is retained line for line, and no part of it is replaced by this reference.

We stress that *we* are not assuming UIP or the K principle as axioms! They are simply properties which a particular type may or may not satisfy (which are equivalent to being a set). Recall from original theorem 3.1.9 that *not* all types are sets.

The following theorem is another useful way to show that types are sets.

Theorem 7.2.2. *Suppose R is a reflexive mere relation on a type X implying identity. Then X is a set, and $R(x, y)$ is equivalent to $x =_X y$ for all $x, y : X$.*

Proof. Let $\rho : \prod_{(x:X)} R(x, x)$ witness reflexivity of R, and let $f : \prod_{(x,y:X)} R(x, y) \to (x =_X y)$ be a witness that R implies identity. Note first that the two statements in the theorem are equivalent. For on one hand, if X is a set, then $x =_X y$

is a mere proposition, and since it is logically equivalent to the mere proposition $R(x, y)$ by hypothesis, it must also be equivalent to it. On the other hand, if $x =_X y$ is equivalent to $R(x, y)$, then like the latter it is a mere proposition for all $x, y : X$, and hence X is a set.

We give two proofs of this theorem. The first shows directly that X is a set; the second shows directly that $R(x, y) \simeq (x = y)$.

First proof: we show that X is a set. The idea is the same as that of original lemma 3.3.4: the function f must be continuous in its arguments x and y. However, it is slightly more notationally complicated because we have to deal with the additional argument of type $R(x, y)$.

Firstly, for any $x : X$ and $p : x =_X x$, consider $\mathsf{apd}_{f(x)}(p)$. This is a dependent path from $f(x, x)$ to itself. Since $f(x, x)$ is still a function $R(x, x) \to (x =_X x)$, by original lemma 2.9.6 this yields for any $r : R(x, x)$ a path

$$p_*(f(x, x, r)) = f(x, x, p_*(r)).$$

On the left-hand side, we have transport in an identity type, which is concatenation. And on the right-hand side, we have $p_*(r) = r$, since both lie in the mere proposition $R(x, x)$. Thus, substituting $r :\equiv \rho(x)$, we obtain

$$f(x, x, \rho(x)) \cdot p = f(x, x, \rho(x)).$$

By cancellation, $p = \mathsf{refl}_x$. So X satisfies Axiom K, and hence is a set.

Second proof: we show that each $f(x, y) : R(x, y) \to x =_X y$ is an equivalence. By original theorem 4.7.7, it suffices to show that f induces an equivalence of total spaces:

$$\Big(\sum_{y:X} R(x, y)\Big) \simeq \Big(\sum_{y:X} x =_X y\Big).$$

By original lemma 3.11.8, the type on the right is contractible, so it suffices to show that the type on the left is contractible. As the center of contraction we take the pair $(x, \rho(x))$. It remains to show, for every $y : X$ and every $H : R(x, y)$ that

$$(x, \rho(x)) = (y, H).$$

But since $R(x, y)$ is a mere proposition, by original theorem 2.7.2 it suffices to show that $x =_X y$, which we get from $f(H)$. □